Project Design Phase
Proposed Solution Template

Date	04 July 2026
Team ID	
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be Solved)	Floods cause significant loss of life, property, crops, and infrastructure due to delayed or inaccurate flood warnings. Existing systems often lack real-time, location-specific predictions, making it difficult for residents and disaster management authorities to take timely preventive actions.
2.	Idea / Solution Description	Develop an intelligent flood prediction system using Machine Learning that analyzes historical flood records, rainfall, weather forecasts, river water levels, soil moisture, and other environmental data to predict flood risks accurately. The system provides real-time, location-based alerts, interactive flood-risk maps, and emergency notifications through web and mobile applications.
3.	Novelty / Uniqueness	• Integrates multiple real-time data sources (weather APIs, river sensors, satellite data).
4.	Social Impact / Customer Satisfaction	• Enhances public safety through early flood warnings. • Reduces loss of lives, property, and agricultural damage. • Improves disaster preparedness and emergency response. • Increases public confidence in flood forecasting systems. • Helps governments and communities make informed decisions during flood emergencies.
5.	Business Model (Revenue Model)	• Government and Disaster Management Department subscriptions. • Software-as-a-Service (SaaS) licensing for municipalities and organizations. • Premium analytics and reporting services for enterprises and insurance companies. • API integration services for third-party applications. • Research collaborations and grants from environmental and disaster management organizations.

6.	Scalability of the Solution	The system is designed using a scalable cloud-based architecture with REST APIs and modular services. It can support thousands of users simultaneously, integrate additional IoT sensors and weather data sources, expand to different regions or countries, and continuously improve prediction accuracy through periodic machine learning model retraining. Cloud deployment enables horizontal scaling, high availability, and reliable disaster monitoring.